# Preprocessing the Polysemous Clitic *-nya* for Indonesian Information Retrieval: A Five-Condition Diagnostic Study on MIRACL-id

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***Abstract*—The Indonesian clitic *-nya* is one of the highest-frequency morphological items in the language and serves at least four distinct functions — possessive, definite, anaphoric, and nominalising — yet standard Indonesian information-retrieval pipelines treat it as a generic suffix to be stripped. We test whether this preprocessing default helps or harms modern retrieval effectiveness through a controlled five-condition study on the Indonesian split of MIRACL: (i) Keep, (ii) Naive regex strip, (iii) dictionary-aware Sastrawi clitic strip, (iv) Sentinel masking, and (v) Rule-based anaphora resolution that rewrites anaphoric *-nya* occurrences with their resolved antecedent using surface-context heuristics and applies to all 1.4M passages at CPU-only cost. We evaluate both sparse (BM25) and dense (BGE-m3) retrievers and report nDCG@10 and MRR@100 across all conditions, with a paired Wilcoxon signed-rank protocol and a Bonferroni-corrected pairwise post-hoc. TODO: Insert headline result once experiments complete. Our findings inform the preprocessing decisions made by every Indonesian retrieval system that currently inherits English-centric defaults and quantify, for the first time, the harm of the naive *-nya* regex that is widely distributed in practitioner code.**

***Index Terms*—information retrieval, Indonesian, clitic, polysemy, dense retrieval, BM25, BGE-m3, MIRACL.**

## I. Introduction

Indonesian, the official language of the world's fourth-most populous country, has roughly 270 million speakers but a comparatively thin natural-language-processing infrastructure. The clitic *-nya* appears in approximately 19.5% of suffix occurrences in Indonesian corpus data [1], making it one of the highest-frequency morphological items in the language. It is also one of the most polysemous: a single surface form encodes at least four distinct grammatical functions: third-person singular possessive (*rumahnya*, 'her house'), definite marker (*rumahnya*, 'the house' given prior reference), anaphoric pronoun referring to a previously mentioned entity (*pidatonya*, 'his/her speech'), and nominaliser (*datangnya*, 'the arrival').

Standard Indonesian information-retrieval pipelines treat *-nya* as a generic suffix to be stripped by a stemmer such as the Nazief–Adriani algorithm [2] or its modern descendant Sastrawi [3]. The most authoritative empirical study on Indonesian stemming and retrieval effectiveness, by Tala [4], predates BERT and dense retrieval entirely. The preprocessing defaults inherited from that pre-neural era propagate into modern retrieval systems built on multilingual dense encoders such as BGE-m3 [5] and multilingual E5 [6], where they have not been controlled-tested. Recent surveys of preprocessing on Transformer-based models report that classical preprocessing often does not help, or actively hurts, modern neural systems on English-centric benchmarks [7]. Whether that finding generalises to a morphologically distinctive low-resource language with a high-frequency polysemous clitic is an open empirical question.

A separate and equally untested concern is the *quality* of the *-nya* stripping itself. A naive regex of the form `(\w+?)nya\b` — which appears in widely-distributed Indonesian NLP tutorials — over-strips a substantial set of high-frequency Indonesian non-clitic words such as *punya* 'to have', *tanya* 'to ask', *hanya* 'only', and *biasanya* 'usually', as well as loanwords and proper nouns including *Kenya*, *Sonya*, and the personal name *Tanya*. Quantifying the harm caused by this naive default is a public-good contribution distinct from the more sophisticated question of whether *any* preprocessing helps modern retrievers.

### A. Research questions

This paper isolates Indonesian *-nya* preprocessing as an experimental variable in modern retrieval and addresses five pre-registered questions on the Indonesian split of MIRACL [8]:

**RQ1 (primary).** Among five preprocessing strategies for the Indonesian polysemous clitic *-nya* — Keep, Naive regex strip, dictionary-aware Sastrawi clitic strip, Sentinel masking, and rule-based anaphora resolution — which yields the highest retrieval effectiveness on MIRACL-id?

**RQ2 (architecture interaction).** Does the optimal preprocessing strategy depend on retriever architecture, specifically between sparse BM25 and dense BGE-m3?

**RQ3 (harm quantification).** How large is the retrieval-effectiveness penalty incurred by the naive regex *-nya* pattern that is widely distributed in practitioner code despite over-stripping common non-clitic words?

**RQ4 (anaphora-resolution gain).** Does a fully automated rule-based anaphora resolver for *-nya* yield a measurable retrieval-effectiveness gain over the best non-resolving strategy, and on which query population is the gain concentrated?

**RQ5 (sensitivity stratification).** Are the differences between strategies larger on queries whose gold passages contain high concentrations of anaphoric *-nya* than on queries with little or no *-nya* content?

### B. Contributions

Our contributions are:

- The first controlled study isolating Indonesian *-nya* preprocessing as an experimental variable in a modern dense-retrieval setting, on both sparse (BM25) and dense (BGE-m3) retrievers.
- Empirical quantification of the harm caused by the naive regex *-nya* stripping pattern that is widely distributed in practitioner code despite over-stripping common non-clitic words — a harm-baseline that, to our knowledge, has not been previously measured.
- A fully automated rule-based *-nya* anaphora resolver that operates over the full MIRACL-id corpus at CPU-only cost, and a measurement of the retrieval-effectiveness gain it produces over the best non-resolving strategy.
- A sensitivity-stratified analysis identifying the query population on which preprocessing decisions matter most, providing actionable guidance for system designers building Indonesian RAG and search products.

The remainder of the paper is organised as follows. Section II situates the work in four adjacent literatures: Indonesian morphology and clitic handling, Indonesian retrieval and stemming, anaphora and coreference for retrieval, and preprocessing for transformer-based models. Section III specifies the five preprocessing conditions and the experimental design. Section IV describes implementation details, and Section V presents results. Section VI interprets the findings and discusses limitations. Section VII concludes.

## II. Related Work

Four bodies of work are directly relevant: the linguistics and computational morphology of Indonesian *-nya*; Indonesian information retrieval and stemming; anaphora and coreference resolution applied to retrieval; and the general question of preprocessing for transformer-based retrievers.

### A. Indonesian morphology and the clitic -nya

*-nya* is descriptively well-documented in the linguistics literature, where it is consistently analysed as polysemous across at least four functions: possessive third-person singular, definite marker, anaphoric pronoun, and nominaliser [1]. Computational morphology tools that handle Indonesian cliticisation include MorphInd [9] and the recent IndoMorph engine [10]. Larasati's earlier treatment of Indonesian clitics in the context of machine translation [11] provides one of the few public datasets distinguishing clitic from non-clitic occurrences of the *-nya* surface form. The most directly relevant computational study of the clitic comes from Malay rather than Indonesian: a recent Malay-clitic classification study [12] classifies possessive *nya* into pleonastic, human-referring, and non-human-referring uses with accuracies of 88%, 94%, and 63% respectively, but stops at classification and does not connect to retrieval.

### B. Indonesian information retrieval and stemming

The canonical study of Indonesian stemming for IR is the study by Tala [4], which evaluates several stemming algorithms on a TREC-style Indonesian collection using pre-neural vector-space methods. The result — that stemming generally helps Indonesian IR — has been widely cited but has not, to our knowledge, been re-validated against modern dense retrieval. Subsequent work compares stemming algorithms (Sastrawi [3], Nazief–Adriani [2], Porter variants) on *stemming accuracy* rather than retrieval effectiveness; Sastrawi consistently reports the highest accuracy at approximately 95.2%. Indonesian language models including IndoBERT [13] and IndoLEM [14] have established benchmarks for Indonesian NLU, and the NusaCrowd consortium [15] aggregates open Indonesian NLP resources. The Indonesian split of MIRACL [8] provides a high-quality, native-annotated retrieval benchmark, and Mr. TyDi [16] additionally provides an Indonesian split designed for dense-retrieval evaluation. None of these resources isolates clitic preprocessing as an experimental variable.

### C. Anaphora, coreference, and retrieval

Mitkov [17] demonstrated in the pre-neural era that anaphora resolution improves both retrieval and question answering for English, and a recent ACL Student Research Workshop paper by Jang et al. [18] extends the finding to modern retrieval-augmented generation, showing that explicit coreference resolution measurably improves both retrieval and downstream generation for English RAG. For Indonesian specifically, Auliarachman and Purwarianti [19] present a CNN-based mention-pair coreference resolver achieving an F1 of approximately 76% with gold-singleton classification, and Saputri et al. [20] present a multi-pass sieve resolver; neither system is connected to retrieval evaluation. The intersection of *-nya*-aware preprocessing and modern Indonesian dense retrieval — the focus of the present study — is, after a deliberate scoping search, the gap that this paper fills.

### D. Preprocessing for transformer-based models

A recent comparative survey by Camacho-Collados et al. [7] reports that classical preprocessing techniques (stemming, stopword removal, lowercasing) often do not benefit transformer-based classifiers in English-centric tasks. The generality of that finding to morphologically distinctive low-resource languages, and specifically to retrieval rather than classification, is open. Our results provide one data point.

## III. Methodology

### A. Experimental design

We adopt a within-subjects factorial design with two factors: preprocessing strategy (five levels) and retriever architecture (two levels). Every dev-set query is evaluated under every combination of conditions.

## B. Hypotheses

We pre-register five falsifiable hypotheses corresponding to RQ1–RQ5:

**H1 (omnibus).** At least one of the five preprocessing strategies yields a significantly different per-query nDCG@10 distribution than the others, separately for each retriever architecture. *Test:* Friedman's test, $\alpha = 0.05$.

**H2 (harm of naive strip).** Naive regex *-nya* stripping yields significantly lower nDCG@10 than Keep on both retrievers, due to over-stripping of high-frequency non-clitic words such as *punya*, *tanya*, and *hanya*. *Test:* one-tailed paired Wilcoxon signed-rank, $\alpha = 0.05$.

**H3 (architecture interaction).** The optimal preprocessing strategy among Keep, Sastrawi clitic, Sentinel, and Rule-resolved differs between BM25 and BGE-m3. *Test:* comparison of best-performing strategy per retriever with bootstrap confidence intervals on the gain difference.

**H4 (anaphora-resolution gain).** Rule-resolved yields significantly higher nDCG@10 than the best of Keep, Sastrawi clitic, and Sentinel for both retrievers. *Test:* paired Wilcoxon signed-rank versus the per-retriever-best non-resolving strategy, Bonferroni-corrected $\alpha = 0.025$.

**H5 (sensitivity stratification).** The magnitude of differences between strategies is larger on high--*nya*-sensitivity queries than on low-sensitivity queries. *Test:* mixed-effects model with strategy-by-stratum interaction term.

## C. Preprocessing conditions

Each strategy is applied identically to queries and passages.

1) **Keep** — identity preprocessing; UTF-8 normalised to NFC. Baseline.
2) **Naive strip** — regex `(\w+?)nya\b` replaced with the captured group. Included as a deliberate harm-baseline; over-strips *punya*, *tanya*, *hanya*, *biasanya*, and proper nouns ending in *-nya*.
3) **Sastrawi clitic** — Sastrawi's `RemovePossessivePronoun` filter applied with an explicit Indonesian-dictionary guard; strips *-nya* only when the result is a known root, protecting non-clitic words.
4) **Sentinel** — *-nya* replaced with a spaced sentinel token <NYA>. Diagnostic condition that removes lexical match while preserving the information that a clitic was present.
5) **Rule-resolved** — rule-based anaphora resolution: every *-nya* occurrence is classified by function (possessive, definite, anaphoric, nominaliser) using surface-context heuristics, and anaphoric occurrences are substituted with the most recent salient antecedent noun phrase from the prior one to two sentences. The full passage corpus is rewritten by this pipeline in CPU minutes, no human annotation and no commercial API. Framed as best-effort automated resolution rather than an oracle ceiling.

Implementation details for each strategy — including dictionary-guard wrapping for Strategy 3, the false-positive verification protocol for Strategy 2, and the heuristic-classifier plus antecedent-selection rules for Strategy 5 — are reported in Section IV.

## D. Retrievers

We evaluate two retriever architectures spanning the sparse–dense spectrum.

**BM25.** Standard BM25 implementation in Pyserini, with $k_1 = 0.9$ and $b = 0.4$ (standard MIRACL configuration). Tokenisation uses the default whitespace-plus-punctuation analyser.

**BGE-m3.** The dense embedding mode of BGE-m3 [5], with maximum sequence length capped at 512 tokens. Indices built using FAISS HNSW with $M = 32$ and `efConstruction` = 200.

## E. Dataset

We evaluate on the Indonesian split of MIRACL [8]: approximately 1.4 million Wikipedia passages and 960 development-set queries with native-annotated relevance judgments.

## F. Metrics

**Primary:** nDCG@10. **Secondary:** MRR@100, nDCG@1, Recall@10, Recall@100. Per-query metrics are computed under every condition; nDCG@1 is reported as a more sensitive metric in case of saturation effects on dense baselines.

## G. Sensitivity stratification

Per-query *-nya* sensitivity scores are computed using the formula

$$\text{sensitivity}(q) = \alpha\, c_{\text{nya}}(q) + \beta\, c_{\text{ana}}(q) + \gamma\, \mathbf{1}[\text{queried entity referenced via} \tag{1}$$

where $c_{\text{nya}}$ and $c_{\text{ana}}$ are counts of total and heuristically-detected anaphoric *-nya* occurrences in the gold-relevant passages, with weights $\alpha = 1$, $\beta = 2$, $\gamma = 3$ (frozen prior to main analysis). Queries are stratified into low/mid/high tertiles for the H5 analysis.

## H. Statistical analysis

Omnibus tests use Friedman's test on per-query nDCG@10. Pairwise post-hoc tests use Wilcoxon signed-rank with Bonferroni correction across the $\binom{5}{2} = 10$ pairs per retriever ($\alpha = 0.005$ within each retriever family). Effect sizes are reported as Cliff's $\delta$. Bootstrap 95% confidence intervals on $\Delta$nDCG@10 are computed with 10,000 resamples. The H5 stratification analysis uses a mixed-effects model with strategy-by-stratum interaction. The protocol follows the recommendations of Dror et al. [21].

## I. Reproducibility and pre-registration

This methodology was committed to a public Git repository prior to running any experimental code, serving as a lightweight pre-registration of the design. All code, intermediate indices, and analysis scripts will be released under an open licence on completion.

## IV. Experiments

### A. Implementation

**TODO:** Fill in once code is written. Cover: pyserini version, sentence-transformers version, PySastrawi version, FAISS configuration, hardware (Kaggle T4), random seeds.

### B. Preprocessing pipeline verification

**TODO:** Day 1 task. Report the false-positive rate of Strategy 2 (Naive strip) on a 50-word test set of known non-clitic words ending in 'nya'. Report the same set after Strategy 3 (Sastrawi clitic with dictionary guard) to confirm protection.

### C. Indexing

For each of the five preprocessing strategies, two indices are built per the BM25 and BGE-m3 configurations specified in Section III.

### D. Rule-based resolver implementation (Strategy 5)

The rule-based *-nya* anaphora resolver is implemented as a three-stage pipeline: (i) clitic detection using the same dictionary-guarded matcher as Strategies 3 and 4, (ii) functional-class classification (possessive, definite, anaphoric, nominaliser) using surface-context heuristics over the host word's part-of-speech and the salience of preceding noun phrases, and (iii) antecedent substitution for occurrences classified as anaphoric, selecting the most recent salient noun phrase from the prior one to two sentences. The pipeline is applied uniformly to all 1.4M MIRACL-id passages. **TODO:** Report total wall-clock time and CPU profile once implementation is complete.

### E. Sensitivity stratification

Per-query *-nya* sensitivity scores are computed using the formula

$$\text{sensitivity}(q) = \alpha\cdot c_{\text{nya}}(q)+\beta\cdot c_{\text{ana}}(q)+\gamma\cdot\mathbf{1}[\text{queried entity referenced via -nya}] \tag{2}$$

where $c_{\text{nya}}$ and $c_{\text{ana}}$ are counts of total and heuristically-detected anaphoric *-nya* occurrences in the gold-relevant passages, with weights $\alpha = 1$, $\beta = 2$, $\gamma = 3$ (frozen prior to main analysis). Queries are stratified into low/mid/high tertiles for the H5 analysis.

## V. Results

**TODO:** Populate after experiments. Suggested structure:

### A. Main results

**TODO:** Table: per-strategy nDCG@10 and MRR@100 for BM25 and BGE-m3, full dev set.

### B. Naive strip is harmful (H2)

**TODO:** Reports paired Wilcoxon, effect size, bootstrap CI.

### C. Architecture-by-strategy interaction (H3)

**TODO:** Compare best-strategy gains across retrievers.

### D. Rule-based resolution gain (H4)

**TODO:** Per-retriever comparison: Rule-resolved vs argmax(Keep, Sastrawi clitic, Sentinel). Paired Wilcoxon with Bonferroni-corrected alpha; report effect size and bootstrap CI.

### E. Sensitivity stratification (H5)

**TODO:** Mixed-effects model output, plus stratified effect-size bars.

### F. Failure analysis

**TODO:** 20 hand-picked queries where Sastrawi clitic and Rule-resolved disagree most — error pattern taxonomy.

## VI. Discussion

**TODO:** Populate after results. Address each of:

### A. What our results say about Indonesian preprocessing defaults

**TODO:** Connect findings back to the Tala (2003) baseline and the Camacho-Collados et al. (2023) finding for English. Position as confirmatory or contrastive.

### B. Implications for production Indonesian retrieval systems

**TODO:** Concrete recommendation: should practitioners preprocess *-nya*, and if so, how? Tie to the architecture-interaction result.

### C. Rule-based anaphora resolution and the ceiling for future research

**TODO:** If Rule-resolved gain is large: an automated -nya resolver is a worthwhile engineering investment for Indonesian retrieval. If small: surface-level handling (Strategies 3 or 4) is sufficient and the bottleneck for retrieval improvement lies elsewhere.

### D. Limitations

Strategy 5 (Rule-resolved) is a best-effort automated resolver; it inherits the error rate of its heuristic classifier and antecedent-selection rules into the retrieval result. A practical upper bound on retrieval gain from *-nya* anaphora resolution would require either a higher-accuracy resolver (neural or LLM-based) or oracle manual annotation. The realistic deployable gain reported here is bounded above by the gain that a perfect resolver would produce, and below by the gain of a degenerate resolver that classifies every occurrence as non-anaphoric.

Our evaluation is on a single benchmark (MIRACL-id). **TODO:** Add note on Mr. TyDi-id replication if attempted. Generalisation to other Indonesian retrieval contexts (legal, medical, social-media) requires additional evaluation.

The Sastrawi clitic condition (Strategy 3) depends on Sastrawi's internal Indonesian root dictionary; rare or technical-domain vocabulary may produce false positives and false negatives. **TODO:** Quantify if observed in failure analysis.

The naive-strip baseline (Strategy 2) is a deliberate harm-baseline; the more important comparison is between Keep, Sastrawi clitic, Sentinel, and Rule-resolved. Naive is included to quantify the harm of a preprocessing pattern that does appear in practitioner code, not to argue that anyone should use it.

### *E. Threats to construct validity*

The choice to apply preprocessing symmetrically to queries and passages reflects standard IR experimental practice but may differ from production deployments where query- and corpus-side preprocessing are independently configured. **TODO:** Discuss if any condition appears unusually sensitive to this choice.

## VII. Conclusion

**TODO:** Populate after results. Suggested structure: one sentence summarising the empirical finding across the five strategies; one sentence on practical recommendation for Indonesian retrieval practitioners; one sentence on the gain from rule-based anaphora resolution and what it implies for future Indonesian anaphora-aware retrieval research; one sentence pointing to released artefacts.

## Reproducibility

All code, processed data, intermediate indices, and analysis scripts are available at https://github.com/shaneemichael/TA_TBI. The methodology was pre-registered via a timestamped commit before experimental code was run; the relevant commit hash is reported in the supplementary materials.

## AI Assistance Disclosure

Literature scanning, code skeleton drafting, and LaTeX scaffolding were assisted by Claude (Sonnet/Opus). All experimental design choices, hypothesis formulation, statistical analyses, interpretation of results, and final manuscript content are the work of the human author.

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**TODO:** Add when paper is complete: dataset providers (MIRACL team), Sastrawi maintainers, any collaborators or reviewers.

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